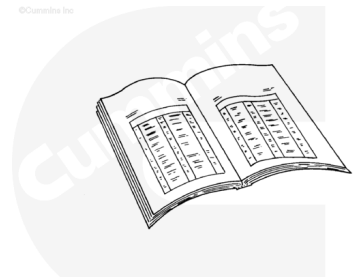


## Trainings ▾

### Preparatory Steps

#### **⚠ WARNING ⚠**

**Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.**



ck800wa

#### **⚠ CAUTION ⚠**

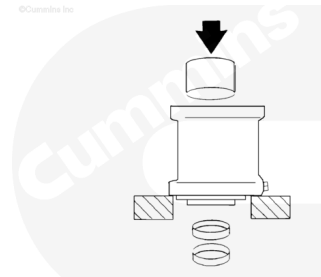
**Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.**

- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Remove the water pump. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html>)
- Remove the belt guard. Refer to Procedure 008-001 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-001.html>)
- Remove the alternator drive pulley. Refer to Procedure 009-010 in Section 9. (</qs3/pubsys2/xml/en/procedures/20/20-009-010-tr.html>)
- Remove and disassemble the water pump drive. Refer to Procedure 009-029 in Section 9. (</qs3/pubsys2/xml/en/procedures/20/20-009-029-tr.html>)

## Disassemble

Support the housing.

Use an arbor press and a mandrel to remove the bushings.



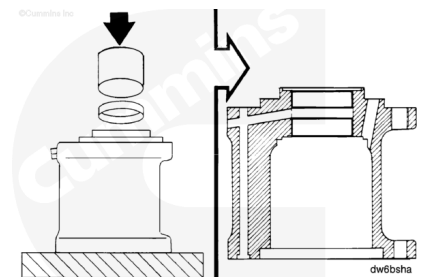
## Assemble

Support the housing.

**⚠ CAUTION ⚠**

**Do not block the oil drilling in the housing.**

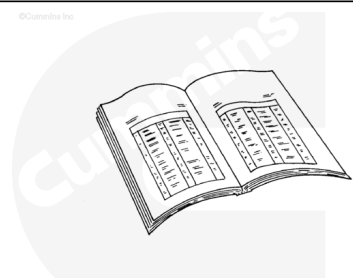
Use an arbor press and a mandrel to install the bushings. The bushing **must** be even or a **maximum** of 0.38 mm [0.015 inch] below the housing surface.



## Finishing Steps

**⚠ CAUTION ⚠**

**Use caution when filling the cooling system so that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is spilled, it must be disposed of in accordance with local environmental regulations.**



- Assemble and install the water pump drive. Refer to Procedure 009-029 in Section 9.  
(/qs3/pubsys2/xml/en/procedures/20/20-009-029-tr.html)
- Install the alternator drive pulley. Refer to Procedure 009-010 in Section 9.  
(/qs3/pubsys2/xml/en/procedures/20/20-009-010-tr.html)

- Install the water pump. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html>)
- Fill with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Install the belt guard. Refer to Procedure 008-001 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-001.html>)
- Operate the engine and check for leaks.

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**Last Modified: 20-Aug-2015**

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